

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

**Product Description:** Puck's G Balanced Salt Solution, with calcium, with magnesium, with phenol red  
**Cat No. :** J67542

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### 1.3. Details of the supplier of the safety data sheet

**Company**  
Avocado Research Chemicals Ltd.  
(Part of Thermo Fisher Scientific)  
Shore Road, Heysham  
Lancashire, LA3 2XY,  
United Kingdom  
Office Tel: +44 (0) 1524 850506  
Office Fax: +44 (0) 1524 850608

**E-mail address** [begin.sdsdesk@thermofisher.com](mailto:begin.sdsdesk@thermofisher.com)

### 1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

#### CLP Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

##### Physical hazards

Based on available data, the classification criteria are not met

##### Health hazards

Based on available data, the classification criteria are not met

##### Environmental hazards

Based on available data, the classification criteria are not met

# SAFETY DATA SHEET

Puck's G Balanced Salt Solution, with calcium, with magnesium, with phenol red

Revision Date 20-Mar-2024

Full text of Hazard Statements: see section 16

## 2.2. Label elements

None required

## 2.3. Other hazards

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.2. Mixtures

| Component                      | CAS No     | EC No             | Weight % | CLP Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567 |
|--------------------------------|------------|-------------------|----------|-----------------------------------------------------------------------------------------|
| Water                          | 7732-18-5  | 231-791-2         | 98.64    | -                                                                                       |
| Sodium chloride                | 7647-14-5  | 231-598-3         | 0.801    | -                                                                                       |
| Potassium chloride             | 7447-40-7  | 231-211-8         | 0.395    | -                                                                                       |
| Glucose                        | 50-99-7    | EEC No. 200-075-1 | 0.11     | -                                                                                       |
| Sodium phosphate dibasic       | 7558-79-4  | 231-448-7         | 0.016    | -                                                                                       |
| Dihydrogen potassium phosphate | 7778-77-0  | 231-913-4         | 0.015    | -                                                                                       |
| Calcium chloride               | 10043-52-4 | 233-140-8         | 0.011    | Eye Irrit. 2 (H319)                                                                     |
| Magnesium chloride             | 7786-30-3  | EEC No. 232-094-6 | 0.006    | -                                                                                       |
| Phenol red                     | 143-74-8   | EEC No. 205-609-7 | 0.004    | Skin Irrit. 2 (H315)<br>Eye Irrit. 2 (H319)<br>STOT SE 3 (H335)                         |

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                           |                                                                                                                         |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.         |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur. |
| <b>Ingestion</b>                          | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                   |
| <b>Inhalation</b>                         | Remove to fresh air. Get medical attention immediately if symptoms occur.                                               |
| <b>Self-Protection of the First Aider</b> | No special precautions required.                                                                                        |

# SAFETY DATA SHEET

Puck's G Balanced Salt Solution, with calcium, with magnesium, with phenol red

Revision Date 20-Mar-2024

## 4.2. Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

## 4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### **Suitable Extinguishing Media**

Not combustible.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### 5.2. Special hazards arising from the substance or mixture

None reasonably foreseeable.

#### **Hazardous Combustion Products**

Sulfur oxides, Hydrogen chloride, Oxides of phosphorus, Calcium oxides, Sodium oxides, Magnesium oxides.

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required.

### 6.2. Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

### 6.3. Methods and material for containment and cleaning up

Sweep up and shovel into suitable containers for disposal.

### 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation.

# SAFETY DATA SHEET

Puck's G Balanced Salt Solution, with calcium, with magnesium, with phenol red

Revision Date 20-Mar-2024

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 7.2. Conditions for safe storage, including any incompatibilities

Keep container tightly closed in a dry and well-ventilated place.

Technical Rules for Hazardous Substances (TRGS) 510

Class 12

Storage Class (LGK) (Germany)

## 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s):

#### Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

#### Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                                 | Acute effects local (Oral) | Acute effects systemic (Oral) | Chronic effects local (Oral) | Chronic effects systemic (Oral) |
|-------------------------------------------|----------------------------|-------------------------------|------------------------------|---------------------------------|
| Magnesium chloride<br>7786-30-3 ( 0.006 ) |                            |                               |                              | 7 mg/kg bw/day                  |

| Component                                 | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|-------------------------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Sodium chloride<br>7647-14-5 ( 0.801 )    |                              | DNEL = 295.52mg/kg bw/day       |                                | DNEL = 295.52mg/kg bw/day         |
| Potassium chloride<br>7447-40-7 ( 0.395 ) |                              | DNEL = 910mg/kg bw/day          |                                | DNEL = 303mg/kg bw/day            |

| Component                                             | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|-------------------------------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Sodium chloride<br>7647-14-5 ( 0.801 )                |                                  | DNEL = 2068.62mg/m <sup>3</sup>     |                                    | DNEL = 2068.62mg/m <sup>3</sup>       |
| Potassium chloride<br>7447-40-7 ( 0.395 )             |                                  | DNEL = 5320mg/m <sup>3</sup>        |                                    | DNEL = 1064mg/m <sup>3</sup>          |
| Dihydrogen potassium phosphate<br>7778-77-0 ( 0.015 ) |                                  |                                     |                                    | DNEL = 14.82mg/m <sup>3</sup>         |

#### Predicted No Effect Concentration (PNEC)

See values below.

# SAFETY DATA SHEET

Puck's G Balanced Salt Solution, with calcium, with magnesium, with phenol red

Revision Date 20-Mar-2024

| Component                                       | Fresh water     | Fresh water sediment          | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)         |
|-------------------------------------------------|-----------------|-------------------------------|--------------------|------------------------------------|----------------------------|
| Sodium chloride<br>7647-14-5 ( 0.801 )          | PNEC = 5mg/L    |                               |                    | PNEC = 500mg/L                     | PNEC = 4.86mg/kg soil dw   |
| Potassium chloride<br>7447-40-7 ( 0.395 )       | PNEC = 0.1mg/L  |                               | PNEC = 1mg/L       | PNEC = 10mg/L                      |                            |
| Sodium phosphate dibasic<br>7558-79-4 ( 0.016 ) | PNEC = 0.05mg/L |                               | PNEC = 0.5mg/L     | PNEC = 50mg/L                      |                            |
| Magnesium chloride<br>7786-30-3 ( 0.006 )       | PNEC = 3.21mg/L | PNEC = 288.9mg/kg sediment dw | PNEC = 5.48mg/L    | PNEC = 90mg/L                      | PNEC = 662.77mg/kg soil dw |

| Component                                       | Marine water     | Marine water sediment         | Marine water intermittent | Food chain | Air |
|-------------------------------------------------|------------------|-------------------------------|---------------------------|------------|-----|
| Potassium chloride<br>7447-40-7 ( 0.395 )       | PNEC = 0.1mg/L   |                               |                           |            |     |
| Sodium phosphate dibasic<br>7558-79-4 ( 0.016 ) | PNEC = 0.005mg/L |                               |                           |            |     |
| Magnesium chloride<br>7786-30-3 ( 0.006 )       | PNEC = 0.32mg/L  | PNEC = 28.89mg/kg sediment dw |                           |            |     |

## 8.2. Exposure controls

### Engineering Measures

None under normal use conditions.

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles) (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Natural rubber | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |
| Nitrile rubber |                                   |                 |             |                       |
| Neoprene       |                                   |                 |             |                       |
| PVC            |                                   |                 |             |                       |

#### Skin and body protection

Long sleeved clothing.

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### Respiratory Protection

No protective equipment is needed under normal use conditions.

#### Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced

**Recommended Filter type:** Particle filter

#### Small scale/Laboratory use

Maintain adequate ventilation

#### Environmental exposure controls

No information available.

# SAFETY DATA SHEET

Puck's G Balanced Salt Solution, with calcium, with magnesium, with phenol red

Revision Date 20-Mar-2024

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                         |                          |                                   |
|-----------------------------------------|--------------------------|-----------------------------------|
| Physical State                          | Liquid                   |                                   |
| Appearance                              | Red                      |                                   |
| Odor                                    | No information available |                                   |
| Odor Threshold                          | No data available        |                                   |
| Melting Point/Range                     | No data available        |                                   |
| Softening Point                         | No data available        |                                   |
| Boiling Point/Range                     | No information available |                                   |
| Flammability (liquid)                   | No data available        |                                   |
| Flammability (solid,gas)                | Not applicable           | Liquid                            |
| Explosion Limits                        | No data available        |                                   |
| Flash Point                             | No information available | Method - No information available |
| Autoignition Temperature                | No data available        |                                   |
| Decomposition Temperature               | No data available        |                                   |
| pH                                      | No information available |                                   |
| Viscosity                               | No data available        |                                   |
| Water Solubility                        | Miscible                 |                                   |
| Solubility in other solvents            | No information available |                                   |
| Partition Coefficient (n-octanol/water) |                          |                                   |
| Vapor Pressure                          | 23 hPa @ 20 °C           |                                   |
| Density / Specific Gravity              | No data available        |                                   |
| Bulk Density                            | Not applicable           | Liquid                            |
| Vapor Density                           | No data available        | (Air = 1.0)                       |
| Particle characteristics                | Not applicable (liquid)  |                                   |

### 9.2. Other information

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

|                          |                               |
|--------------------------|-------------------------------|
| Hazardous Polymerization | No information available.     |
| Hazardous Reactions      | None under normal processing. |

### 10.4. Conditions to avoid

Incompatible products. Excess heat.

### 10.5. Incompatible materials

None known.

### 10.6. Hazardous decomposition products

Sulfur oxides. Hydrogen chloride. Oxides of phosphorus. Calcium oxides. Sodium oxides. Magnesium oxides.

# SAFETY DATA SHEET

Puck's G Balanced Salt Solution, with calcium, with magnesium, with phenol red

Revision Date 20-Mar-2024

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

##### (a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

#### Toxicology data for the components

| Component                      | LD50 Oral                 | LD50 Dermal                   | LC50 Inhalation              |
|--------------------------------|---------------------------|-------------------------------|------------------------------|
| Water                          | -                         | -                             | -                            |
| Sodium chloride                | LD50 = 3 g/kg ( Rat )     | LD50 > 10000 mg/kg ( Rabbit ) | LC50 > 42 mg/L ( Rat ) 1 h   |
| Potassium chloride             | LD50 = 2600 mg/kg ( Rat ) | -                             | -                            |
| Glucose                        | 25.8 g/kg ( Rat )         | -                             | -                            |
| Sodium phosphate dibasic       | LD50 = 17 g/kg ( Rat )    | -                             | -                            |
| Dihydrogen potassium phosphate | LD50 = 3200 mg/kg ( Rat ) | LD50 > 4640 mg/kg ( Rabbit )  | LC50 > 0.83 mg/L ( Rat ) 4 h |
| Calcium chloride               | 2301 mg/kg ( Rat )        | LD50 > 5000 mg/kg ( Rabbit )  | -                            |
| Magnesium chloride             | LD50 = 2800 mg/kg ( Rat ) | LD50 > 2000 mg/kg ( Rat )     | -                            |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

##### (d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs

No information available.

(j) aspiration hazard; No data available

Symptoms / effects, both acute and No information available.

# SAFETY DATA SHEET

Puck's G Balanced Salt Solution, with calcium, with magnesium, with phenol red

Revision Date 20-Mar-2024

delayed

## 11.2. Information on other hazards

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

**Ecotoxicity effects** Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

| Component          | Freshwater Fish                                                                              | Water Flea           | Freshwater Algae    |
|--------------------|----------------------------------------------------------------------------------------------|----------------------|---------------------|
| Sodium chloride    | Pimephals prome: LC50: 7650 mg/L/96h                                                         | EC50: 1000 mg/L/48h  |                     |
| Potassium chloride | Lepomis macrochirus: LC50: 1060 mg/L /96h<br>Pimephales promelas: LC50: 750 - 1020 mg/L /96h | EC50: 825 mg/L/48h   | EC50: 2500 mg/L/72h |
| Calcium chloride   | Lepomis macrochirus: LC50: 10650 mg/L/96h                                                    | EC50: 52 mg/L/48h    |                     |
| Magnesium chloride | Pimephales promelas: EC50: 2.12 g/L:96H                                                      | EC50 : 1400 mg/L/24h | EC50: 2200 mg/L/72h |

| Component          | Microtox                                                                                                                                            | M-Factor |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Magnesium chloride | EC50 Pseudomonas putida: EC50:26,14 g/L/h<br>Photobacterium phosphoreum: EC50: 36,3 mg/L/30 min<br>Photobacterium phosphoreum: EC50: 77,2 mg/L/24 h |          |

### 12.2. Persistence and degradability

**Persistence** Miscible with water, Persistence is unlikely, based on information available.

**12.3. Bioaccumulative potential** Bioaccumulation is unlikely

**12.4. Mobility in soil** The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

**12.5. Results of PBT and vPvB assessment** No data available for assessment.

### 12.6. Endocrine disrupting properties

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**12.7. Other adverse effects**  
**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS



# SAFETY DATA SHEET

Puck's G Balanced Salt Solution, with calcium, with magnesium, with phenol red

Revision Date 20-Mar-2024

## 13.1. Waste treatment methods

|                                            |                                                                                                                                                                                                                             |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification. |
| <b>Contaminated Packaging</b>              | Empty remaining contents. Dispose of in accordance with local regulations. Do not re-use empty containers.                                                                                                                  |
| <b>European Waste Catalogue (EWC)</b>      | According to the European Waste Catalog, Waste Codes are not product specific, but application specific.                                                                                                                    |
| <b>Other Information</b>                   | Waste codes should be assigned by the user based on the application for which the product was used.                                                                                                                         |

## SECTION 14: TRANSPORT INFORMATION

**IMDG/IMO** Not regulated

**14.1. UN number**  
**14.2. UN proper shipping name**  
**14.3. Transport hazard class(es)**  
**14.4. Packing group**

**ADR** Not regulated

**14.1. UN number**  
**14.2. UN proper shipping name**  
**14.3. Transport hazard class(es)**  
**14.4. Packing group**

**IATA** Not regulated

**14.1. UN number**  
**14.2. UN proper shipping name**  
**14.3. Transport hazard class(es)**  
**14.4. Packing group**

**14.5. Environmental hazards** No hazards identified

**14.6. Special precautions for user** No special precautions required.

**14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component | CAS No | EINECS | ELINCS | NLP | IECSC | TCSI | KECL | ENCS | ISHL |
|-----------|--------|--------|--------|-----|-------|------|------|------|------|
|-----------|--------|--------|--------|-----|-------|------|------|------|------|

# SAFETY DATA SHEET

Puck's G Balanced Salt Solution, with calcium, with magnesium, with phenol red

Revision Date 20-Mar-2024

|                                |            |           |   |   |   |   |          |   |   |
|--------------------------------|------------|-----------|---|---|---|---|----------|---|---|
| Water                          | 7732-18-5  | 231-791-2 | - | - | X | X | KE-35400 | X | - |
| Sodium chloride                | 7647-14-5  | 231-598-3 | - | - | X | X | KE-31387 | X | X |
| Potassium chloride             | 7447-40-7  | 231-211-8 | - | - | X | X | KE-29086 | X | X |
| Glucose                        | 50-99-7    | 200-075-1 | - | - | X | X | KE-17727 | X | X |
| Sodium phosphate dibasic       | 7558-79-4  | 231-448-7 | - | - | X | X | KE-12344 | X | X |
| Dihydrogen potassium phosphate | 7778-77-0  | 231-913-4 | - | - | X | X | KE-28622 | X | X |
| Calcium chloride               | 10043-52-4 | 233-140-8 | - | - | X | X | KE-04496 | X | X |
| Magnesium chloride             | 7786-30-3  | 232-094-6 | - | - | X | X | KE-22691 | X | X |
| Phenol red                     | 143-74-8   | 205-609-7 | - | - | X | X | KE-02748 | X | X |

| Component                      | CAS No     | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|--------------------------------|------------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Water                          | 7732-18-5  | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Sodium chloride                | 7647-14-5  | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Potassium chloride             | 7447-40-7  | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Glucose                        | 50-99-7    | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Sodium phosphate dibasic       | 7558-79-4  | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Dihydrogen potassium phosphate | 7778-77-0  | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Calcium chloride               | 10043-52-4 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Magnesium chloride             | 7786-30-3  | X    | ACTIVE                                        | X   | -    | X    | X     | X     |
| Phenol red                     | 143-74-8   | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Authorisation/Restrictions according to EU REACH

| Component                      | CAS No     | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|--------------------------------|------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Water                          | 7732-18-5  | -                                                                   | -                                                                             | -                                                                                                     |
| Sodium chloride                | 7647-14-5  | -                                                                   | -                                                                             | -                                                                                                     |
| Potassium chloride             | 7447-40-7  | -                                                                   | -                                                                             | -                                                                                                     |
| Glucose                        | 50-99-7    | -                                                                   | -                                                                             | -                                                                                                     |
| Sodium phosphate dibasic       | 7558-79-4  | -                                                                   | -                                                                             | -                                                                                                     |
| Dihydrogen potassium phosphate | 7778-77-0  | -                                                                   | -                                                                             | -                                                                                                     |
| Calcium chloride               | 10043-52-4 | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)            | -                                                                                                     |
| Magnesium chloride             | 7786-30-3  | -                                                                   | -                                                                             | -                                                                                                     |
| Phenol red                     | 143-74-8   | -                                                                   | -                                                                             | -                                                                                                     |

## REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Seveso III Directive (2012/18/EC)

| Component                      | CAS No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|--------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Water                          | 7732-18-5  | Not applicable                                                                            | Not applicable                                                                           |
| Sodium chloride                | 7647-14-5  | Not applicable                                                                            | Not applicable                                                                           |
| Potassium chloride             | 7447-40-7  | Not applicable                                                                            | Not applicable                                                                           |
| Glucose                        | 50-99-7    | Not applicable                                                                            | Not applicable                                                                           |
| Sodium phosphate dibasic       | 7558-79-4  | Not applicable                                                                            | Not applicable                                                                           |
| Dihydrogen potassium phosphate | 7778-77-0  | Not applicable                                                                            | Not applicable                                                                           |
| Calcium chloride               | 10043-52-4 | Not applicable                                                                            | Not applicable                                                                           |
| Magnesium chloride             | 7786-30-3  | Not applicable                                                                            | Not applicable                                                                           |
| Phenol red                     | 143-74-8   | Not applicable                                                                            | Not applicable                                                                           |

# SAFETY DATA SHEET

Puck's G Balanced Salt Solution, with calcium, with magnesium, with phenol red

Revision Date 20-Mar-2024

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

## National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

Water endangering class = non-hazardous to waters (self classification)

| Component                      | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|--------------------------------|---------------------------------------|-------------------------|
| Sodium chloride                | WGK1                                  |                         |
| Potassium chloride             | WGK1                                  |                         |
| Glucose                        | WGK1                                  |                         |
| Sodium phosphate dibasic       | WGK1                                  |                         |
| Dihydrogen potassium phosphate | WGK1                                  |                         |
| Calcium chloride               | WGK1                                  |                         |
| Magnesium chloride             | WGK1                                  |                         |

| Component          | France - INRS (Tables of occupational diseases)      |
|--------------------|------------------------------------------------------|
| Sodium chloride    | Tableaux des maladies professionnelles (TMP) - RG 78 |
| Potassium chloride | Tableaux des maladies professionnelles (TMP) - RG 67 |

| Component                                 | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Sodium chloride<br>7647-14-5 ( 0.801 )    | Prohibited and Restricted Substances                                                                           |                                                                                 |                                                                                             |
| Calcium chloride<br>10043-52-4 ( 0.011 )  | Prohibited and Restricted Substances                                                                           |                                                                                 |                                                                                             |
| Magnesium chloride<br>7786-30-3 ( 0.006 ) | Prohibited and Restricted Substances                                                                           |                                                                                 |                                                                                             |
| Phenol red<br>143-74-8 ( 0.004 )          | Prohibited and Restricted Substances                                                                           |                                                                                 |                                                                                             |

## 15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H315 - Causes skin irritation

H319 - Causes serious eye irritation

# SAFETY DATA SHEET

Puck's G Balanced Salt Solution, with calcium, with magnesium, with phenol red

Revision Date 20-Mar-2024

H335 - May cause respiratory irritation

## Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadviser - LOLI, Merck index, RTECS

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

**Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:**

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

## **Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

**Prepared By** Health, Safety and Environmental Department

**Revision Date** 20-Mar-2024

**Revision Summary** New emergency telephone response service provider.

**This safety data sheet complies with Regulation UK SI 2019/758 and UK SI 2020/1577 as amended.**

## **Disclaimer**

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**End of Safety Data Sheet**